PROBLEM 14 | Problems T2_CrystStruct.pdf

Crystalline Structures & Miller Indices

XRD Diffraction and Interplanar Spacings in BCC and FCC Iron

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14. Fe has a BCC or an FCC structure, depending on the temperature. The X-ray diffraction (XRD) technique allows determining the spacing or distances between crystalline planes, from which the lattice parameters are deduced. For the BCC structure, the lattice parameter is $a = 0.2864$ Å**,** ***while for the FCC*** $a = 0.3592$ Å**.** *(Note: the official text labels these values in angstroms, but* 0.2864 Å *would be smaller than an atom. They are nanometres:* $a_{\alpha\text{-Fe}} = 0.2864$ nm $=$ 2.864 Å *and* $a_{\gamma\text{-Fe}} = 0.3592$ nm $= 3.592$ Å*. The official answers below carry the same unit slip, i.e. they read* 0.143 nm*,* 0.179 nm*,* 0.2025 nm*,* 0.207 nm*. This solution works in nm.)* ***a)*** *What distance can be expected between the planes* (020) *for both structures?* ***b)*** *Calculate the distance between the most compact planes of both structures. (Solution: a)* $d_{020}(\text{BCC}) = 0.143$ Å *;* $d_{020}(\text{FCC}) = 0.179$ Å*; b)* $\text{BCC}\ (110) = 0.2025$ Å*;* $\text{FCC}\ (111) = 0.207$ Å*)*

Solution: a) $d_{020}(\text{BCC}) = 0.143$ Å**;** $d_{020}(\text{FCC}) = 0.179$ Å**; b)** $\text{BCC}\ (110) = 0.2025$ Å**;** $\text{FCC}\ (111) = 0.207$ Å

PHASE 1: HYPOTHESIS & PARAMETERS

Input Data:

- Lattice parameter of the BCC phase (α-Fe): $a_{\text{BCC}} = 0.2864$ nm.
- Lattice parameter of the FCC phase (γ-Fe): $a_{\text{FCC}} = 0.3592$ nm.
- Planes of Maximum Packing [Session 4 Slide 10]:
- In the BCC structure: densest planes $\{110\}$.
- In the FCC structure: densest planes $\{111\}$.

PHASE 2: FUNDAMENTAL PHYSICS & FORMULA SELECTION

For any crystal with cubic symmetry, the interplanar spacing d_{hkl} separating two parallel planes of the same Miller family is given by the fundamental relation [Session 3 Slide 53]:

$$d_{hkl} = \frac{a}{\sqrt{h^2 + k^2 + l^2}}$$

- For the (020) plane:

$$d_{020} = \frac{a}{\sqrt{0^2 + 2^2 + 0^2}} = \frac{a}{2}$$

- For the most densely packed plane of BCC (110):

$$d_{110}^{\text{BCC}} = \frac{a_{\text{BCC}}}{\sqrt{1^2 + 1^2 + 0^2}} = \frac{a_{\text{BCC}}}{\sqrt{2}}$$

- For the most densely packed plane of FCC (111):

$$d_{111}^{\text{FCC}} = \frac{a_{\text{FCC}}}{\sqrt{1^2 + 1^2 + 1^2}} = \frac{a_{\text{FCC}}}{\sqrt{3}}$$

PHASE 3: ANALYTICAL DERIVATION & CALCULATION STEP-BY-STEP

1. Part a: Distance between Planes (020)

- **For the BCC Structure:**

$$d_{020}(\text{BCC}) = \frac{a_{\text{BCC}}}{2} = \frac{0.2864\ \text{nm}}{2} = \mathbf{0.1432}\ \text{nm} \approx \mathbf{0.143}\ \text{nm}$$

- **For the FCC Structure:**

$$d_{020}(\text{FCC}) = \frac{a_{\text{FCC}}}{2} = \frac{0.3592\ \text{nm}}{2} = \mathbf{0.1796}\ \text{nm} \approx \mathbf{0.180}\ \text{nm}$$

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2. Part b: Distance between the Most Densely Packed Planes

- **BCC Structure ((110) plane):**

$$d_{110}(\text{BCC}) = \frac{a_{\text{BCC}}}{\sqrt{2}} = \frac{0.2864\ \text{nm}}{1.4142136} = \mathbf{0.20252}\ \text{nm} \approx \mathbf{0.2025}\ \text{nm}$$

- **FCC Structure ((111) plane):**

$$d_{111}(\text{FCC}) = \frac{a_{\text{FCC}}}{\sqrt{3}} = \frac{0.3592\ \text{nm}}{1.7320508} = \mathbf{0.20738}\ \text{nm} \approx \mathbf{0.207}\ \text{nm}$$

`PHASE 4: PHYSICAL INTERPRETATION & VERIFICATION`

1. **Correlation with Bragg's Law:** In an X-ray diffraction test with copper $K\alpha$ radiation ($\lambda = 1.5418\ \text{Å} = 0.15418\ \text{nm}$), the diffraction angle θ is inversely related to d_{hkl}:

$$\sin\theta = \frac{\lambda}{2d_{hkl}}$$

Since the most densely packed planes have the **largest interplanar spacing** d_{hkl} ($d_{110}^{\text{BCC}} = 0.2025\ \text{nm}$, $d_{111}^{\text{FCC}} = 0.207\ \text{nm}$), they produce the **first diffraction peak at the smallest angle** 2θ and with the highest intensity in the XRD spectrum.

Bragg angles (Cu $K\alpha$, $\lambda = 0.15418\ \text{nm}$, first order; λ and d must be in the same unit):

$$\sin\theta_{110}^{\text{BCC}} = \frac{0.15418\ \text{nm}}{2 \times 0.20252\ \text{nm}} = 0.3807 \implies \theta = 22.4^\circ\ (2\theta = 44.7^\circ)$$

$$\sin\theta_{111}^{\text{FCC}} = \frac{0.15418\ \text{nm}}{2 \times 0.20738\ \text{nm}} = 0.3717 \implies \theta = 21.8^\circ\ (2\theta = 43.6^\circ)$$

Mixing $\lambda = 1.5418\ \text{Å}$ with d in nm (or the official mislabelled Å values) would give the impossible $\sin\theta = 1.5418/(2 \times 0.2025) = 3.8$.

2. **Very similar spacings:** It is notable that $d_{110}^{\text{BCC}} \approx 0.203\ \text{nm}$ and $d_{111}^{\text{FCC}} \approx 0.207\ \text{nm}$ are practically identical (difference $\approx 2.4\%$), which confirms that the distances between the densest layers of iron atoms hardly change on crossing the allotropic transformation temperature of 910°C.